

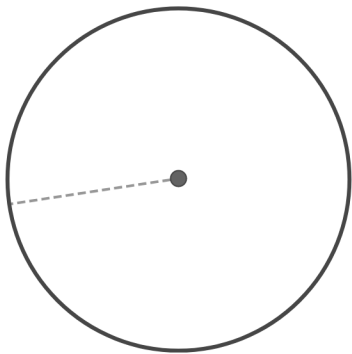
Mr. Davies' Class: homework task

Due date

Tue 27 Apr 10:00pm

1. Identify the highlighted part of each circle below:

a

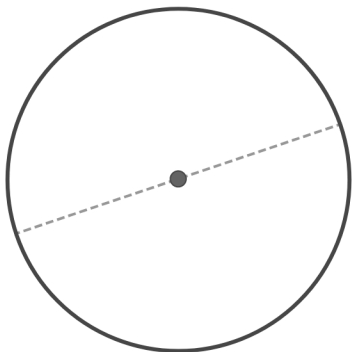


Radius A

Circumference B

Diameter C

b

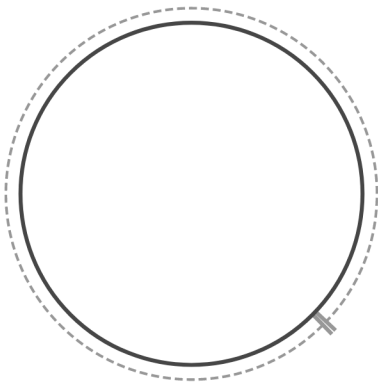


Circumference A

Diameter B

Radius C

c



Diameter

A

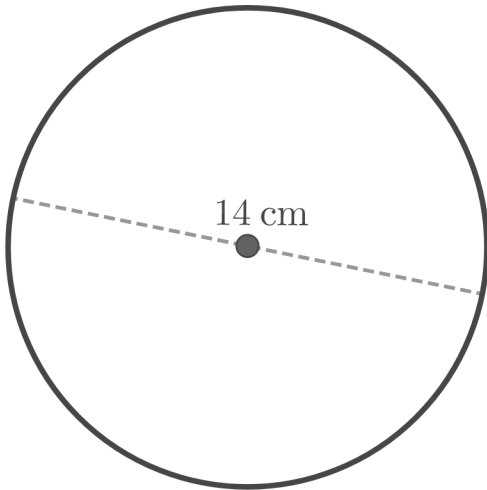
Circumference

B

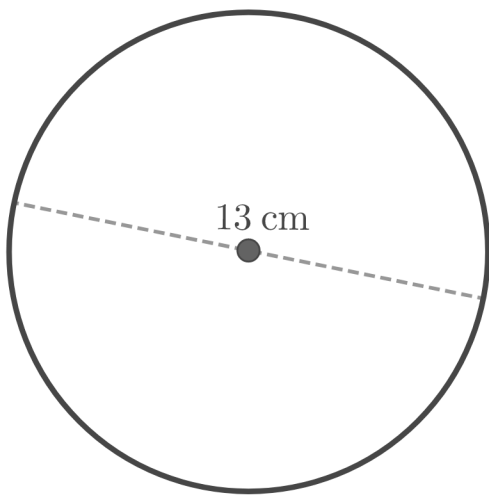
Radius

C

-
2. What is the radius of this circle?



-
3. The formula for the circumference of a circle is:
 $C = \pi d$
where C is the circumference and d is the diameter.
Consider the circle below:

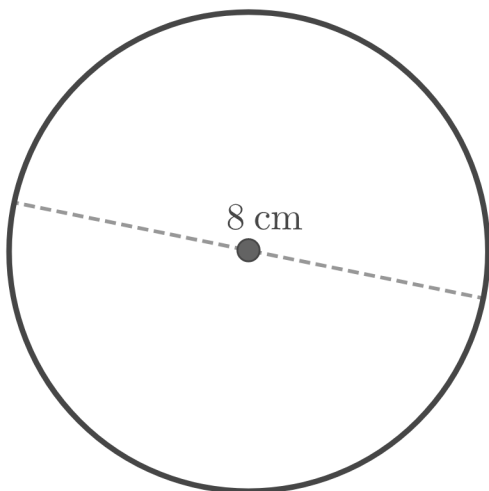


- a What is the exact circumference?
 - b What is the circumference rounded to two decimal places?
-

4. A circle has a radius of 12 mm.

- a What is the exact area of the circle?
 - b What is the area of the circle rounded to two decimal places?
-

5. Consider the circle below:



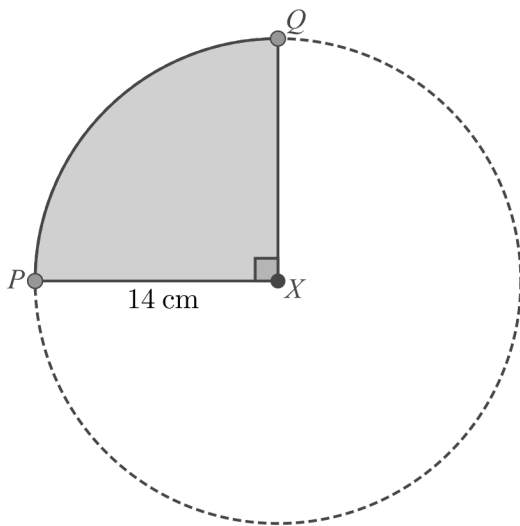
- a What is the radius of the circle?

- b What is the exact area of the circle?
-

6. A circle has a diameter of 22 mm.

- a What is the radius of the circle?
- b What is the exact area of the circle?
- c What is the area rounded to two decimal places?
-

7. The sector in the diagram has an angle of 90° and a radius of 14 cm.



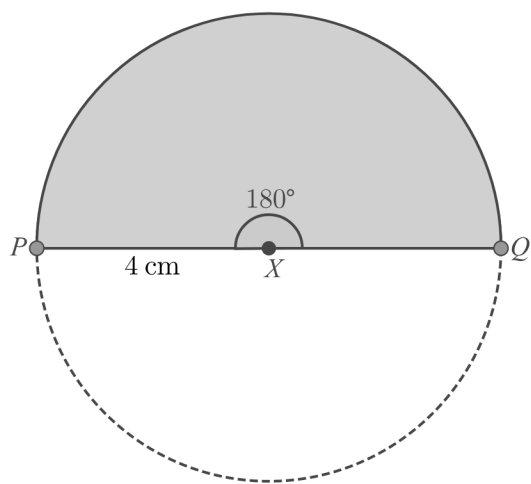
- a What fraction of the circle's area is covered by this sector?
- b Find the exact length of the arc PQ .
- c Fill in the blanks to find the perimeter of the sector, rounded to two decimal places.

Perimeter = Length of arc PQ + length of PX + length of QX

Perimeter = + + (Substitute the exact values for each length)

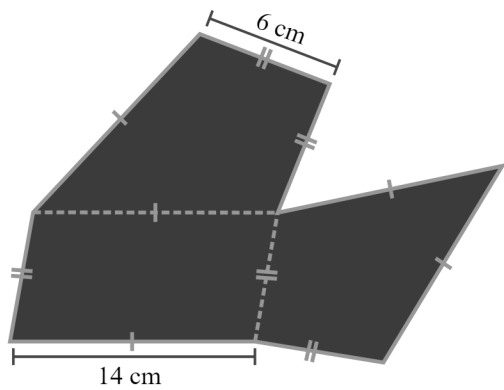
Perimeter = cm (Evaluate, rounding to two decimal places.)

8. The sector in the diagram is equal to half of a circle with a radius of 4 cm:



Find the exact area of the sector.

9. Consider the composite shape.



a What basic shapes make up this composite shape?

- One rhombus and two squares

A
- One parallelogram and two squares

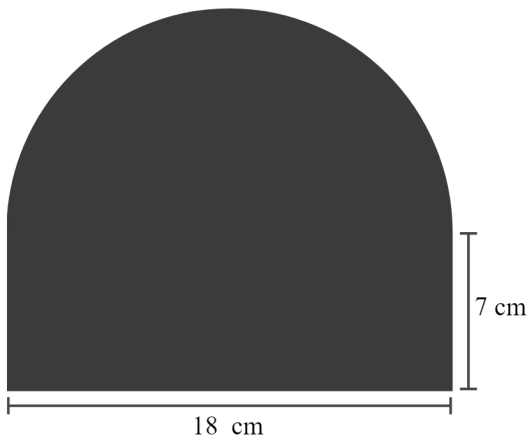
B
- One parallelogram and two kites

C
- One rhombus and two kites

D

b Find the perimeter of the composite shape.

10. Consider the composite shape.



a Which basic shapes make up this composite shape?

One semicircle and one square

A

One circle and one rectangle

B

One semicircle and one rectangle

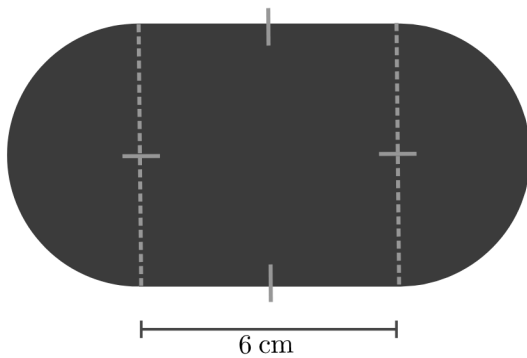
C

One circle and one square

D

b Find the exact perimeter of the composite shape.

11. Consider the composite shape.



a Find the exact perimeter of the composite shape.

b Find the perimeter of the composite shape, rounded to two decimal places.